

A T-I - Analytical Problems

- 34 How a queue may be maintained by a circular array with $\text{MaxSize} = 6$. observe that queue always occupies consecutive and at the end of the array if the queue is viewed as a circular array this means that it still occupies consecutive locations. The queue is empty when $\text{count} = 0$; perform the following Enqueue and Dequeue operations.
- a) Initially queue is empty b) A, B, C are Enqueued ; c) A is Dequeue d) D, E, F are Enqueued ; e) B & C are Dequeued ; f) G is Enqueued ; g) D & E are Dequeued h) H & I are Enqueued ; i) F is Dequeued j) G & H are Dequeued k) I is Dequeued
- Demonstrate with neat sketch of Front rear and count in a queue and circular queue

Step	Operations	Queue (index 0-5)	Front	Rear	Count
a	Empty	(- - - - -)	0	-1	0
b	Enqueue ABC	(A, B, C, - - -)	0	2	3
c	Dequeue A	(- , B, C, - - -)	1	2	2
d	Enqueue D, E, F	(- B, C, D, E, F)	1	5	5
e	Dequeue B, C	(- - , - D, E, F)	3	5	3
f	Enqueue G	(G, - - , D, E, F)	3	0	4
g	Dequeue D, E	(G, - - - - F)	5	0	2
h	Enqueue H, I	(G, H, I, - - F)	5	2	4
i	Dequeue F	(- - - , I, - - -)	0	2	3
j	Dequeue G, H	(- - - - I, - - -)	2	2	1
k	Dequeue I	(- - - - -)	3	2	0

Final observation

Queue becomes empty because $\text{Count} = 0$
Circular queue wraps around from index 5 $\rightarrow$ 0
allowing efficient use of all array location

a) After Enqueue A, B, C

Index : 0 1 2 3 4 5

A

B

C

-

-

-

Front

^
Rear

Count = 3

b) After Enqueue D (wrap Around)

Index : 0 1 2 3 4 5

G

-

-

D

~~E~~

F

^
Rear

Front

Count = 4

c) Final Queue

Index : 0 1 2 3 4 5

-

-

-

-

-

-

Front = 3

Rear = 2

Count = 0 (Queue empty)

Show the linked list configuration resulting from each series of list operations using the List ADT. Assume that List L1 and L2 are empty at the beginning of each series. Show where the current position is in the list. (a) L1.append(10);

```

L1.append(20) ;    L1.append(15) ;    b L2.append(10) ;
L2.append(20) ;    L2.append(15) ;    L2 move To start () ;
L2 insert(39) ;    L2.next() ;    L2 insert(12) ;

```

a) breathations

```
l1.append(10);  
l1.append(20);  
l1.append(15);
```

linked list

Head $\rightarrow 10 \rightarrow 20 \rightarrow 15 \rightarrow \text{null}$
^
current

b) operations

```
L2.append(10)
L2.append(20)
L2.append(15);
L2.move to start();
L2.insert(30)
L2.next();
L2.insert(12);
```

step 1

10 $\rightarrow$ 20 - 15

STEP-2 move to start ()

current
↓
10 → 20 → 15

Step-3: insert (39)

39 → 10 → 20 → 15

↑
current

Step-4 Next (2)

39 → 10 → 20 → 15

↑
current

Step-5 Insert (12)

39 → 12 → 10 → 20 → 15 → NULL

↑
current

Final Answer

Q₁ Final Queue

Front = 3

Rear = 2

Count = 0 (Queue empty)

Q₂ Final Linked Lists

L₁

Head → 10 → 20 → 15 → NULL

L₂

Head → 39 → 12 → 10 → 20 → 15 → NULL